Toward Parameter-Efficient and Trustworthy Transformers for Stress Detection in Reddit Posts

***Abstract*—Social media provides a rich source of data in which to explore large-scale language patterns using automated stress detection. However, the effectiveness of adaptation techniques, model inaccuracies and reliability of model confidence should be explored. Stress classification on Dreaddit is assessed using classical feature representations; fixed pretrained transformer embeddings; training only the classifier head; full fine-tuning; and parameter-efficient adaptation (LoRA) with MiniLM, MPNet and DistilBERT. Misclassifications are explored via error analysis and LIME. Calibration and uncertainty are explored via confidence-based selective prediction and temperature scaling. Fixed pretrained embeddings outperform Word2Vec baseline models and adaptation also provides an advantage. A matched ten-seed experiment compared full fine-tuning and MPNet LoRA-large at equal effective batch sizes. MPNet LoRA-large achieved a mean ROC-AUC of 0.913 $\pm$ 0.003 and mean F1-score of 0.847 $\pm$ 0.005 with about 1.07% of total parameters being updated during training. Full fine-tuning produced a mean ROC-AUC of 0.894 $\pm$ 0.016 and mean F1-score of 0.822 $\pm$ 0.011. In the original selected-model analysis, there were approximately 2.20 times more false positives than false negatives. There is also evidence of repetition of self-referential and affective vocabulary in the misclassified posts. In the original selected LoRA-large model, temperature scaling reduced ECE from 0.0887 to 0.0343; the matched ten-seed mean decreased from 0.1003 to 0.0434. Using confidence-based selective prediction, there is an increase in accuracy on retained samples as fewer samples are considered. Overall, an integration of parameter-efficient adaptation techniques with reliability, explainability and error analysis can yield effective stress classification. An anonymized reproducibility artifact is available at https://github.com/nlp-artifact-review/stress-nlp-artifact-review.**

***Index Terms*—deep learning, machine learning, affective computing, social media analysis.**

I. Introduction

Between 2009 and 2025, the percentage of employees experiencing daily stress increased overall from 31% to 40% worldwide, a notable rise in less than two decades that reflects the substantial prevalence of stress today [1]. This increase, however, extends beyond the workplace, with global stress among adults rising overall from 30% in 2014 to 37% in 2024 [2]. Figure 1 demonstrates how reported stress has changed over time among adults and employees worldwide from 2014–2024.

While stress is a normal physiological and psychological response to external challenges, both brief naturalistic stressors and chronic stressors can produce deleterious effects for those who experience them [3]–[5]. In the short term, stress can be a prominent disruptor of daily activities: it can lead to impaired working memory, reduced cognitive flexibility, poorer decision-making, and disrupted sleep [6]–[8]. Beyond these immediate effects, higher perceived stress correlates with substantially

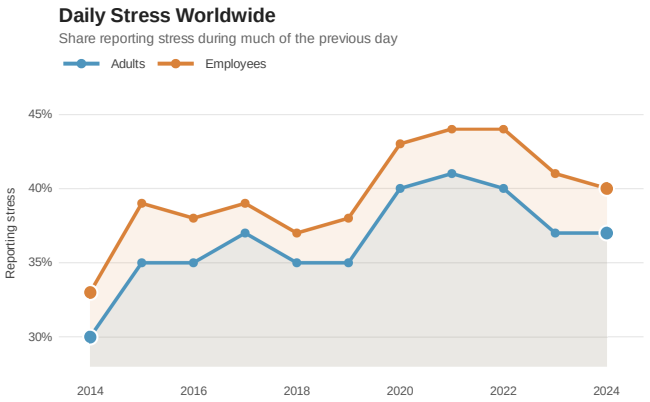


Fig. 1: Global trends in reported stress among adults and employees worldwide, 2014–2024. Data from [1], [2].

greater rates of depression and has also been linked to higher mortality risks and a higher risk of coronary heart disease [9]–[11]. Its global prevalence and effects on immediate functioning and long-term health make the topic of stress an increasingly important subject of study.

Social media platforms have become major communication channels for a substantial share of the global population, and perceived stress has been shown to manifest in the linguistic patterns of users on these platforms—patterns that include greater use of self-focused language and more frequent discussion of exhaustion and pain [12], [13]. The large amounts of text being shared through social media platforms every day provide an abundance of data from which stress-related insights can be obtained. However, the volume of this data and the velocity at which it is generated make comprehensive manual analysis infeasible [14]. Through computational analysis of text data, linguistic patterns associated with psychological states such as stress can be automatically identified, streamlining an otherwise impractical process [13], [15]. Recent studies have demonstrated this across related psychological and mental-health tasks, such as analysis of Reddit and Twitter posts to classify depression and anxiety, and clinical interview transcripts for depression detection [16]–[18]. Among the online sources used for these studies, Reddit is particularly well suited for this type of research because of its comparatively longer-form posts, large number of communities, and continuously generated content that provides a rich source of linguistic data across different contexts [19], [20]. The Dreaddit dataset provides a structured way to study this form of content by enabling multi-domain modeling of stress across Reddit communities [19]. Its

labeled text allows computational approaches to be evaluated on their ability to identify stress-related linguistic features and expressions of stress across different contexts. Hence, there is value in examining how different computational models identify stress in Reddit text and the factors that shape their predictions.

Existing natural language processing and machine learning approaches can identify psychological and mental health signals in social media text, and the increasing sophistication of pretrained models has further expanded their ability to identify linguistic patterns within such texts [16]–[18]. Currently, many studies place an emphasis on the predictive performance of a model, though understanding why particular predictions occur remains important, especially for mental health applications, motivating greater attention to interpretability and error analysis [21]–[23]. Recent mental health NLP research has therefore increasingly incorporated more interpretable approaches, such as symptom-based interpretation and clinically grounded explanations, reflecting a move beyond purely black-box classification; despite this progress, interpretability remains a relevant consideration when evaluating models [21]–[23]. Furthermore, models may rely heavily on superficial lexical cues or other spurious patterns, leading to predictions based on the appearance of particular words or expressions while failing to fully account for the broader context in which those features appear [24]. While certain tokens may be indicative of stress-related topics, reliance on such surface-level patterns raises the possibility that models may conflate discussions of stress with expressions of stress. Beyond model-based limitations, computational efficiency remains a notable factor when adapting pretrained language models: full fine-tuning requires updating all model parameters, which can lead to substantial computational costs, training time, and resource requirements [25]. This computational burden motivates comparing standard fine-tuning with parameter-efficient methods such as LoRA [25].

This paper investigates automated stress detection using Reddit text from the Dreaddit dataset [19], examining approaches across several levels of modeling complexity to investigate the aforementioned limitations within the field. The study uses human-interpretable text analyses to examine observable linguistic characteristics of stress and non-stress posts through unigrams, bigrams, TF-IDF, and readability, while predictive modeling uses Word2Vec features [26] with Random Forest [27] and XGBoost [28], as well as pretrained embeddings from MiniLM [29], MPNet [30], and DistilBERT [31]. Additionally, the study compares standard fine-tuning with LoRA [25] in terms of the number of trainable parameters and predictive performance on the stress prediction task. False positives and false negatives were analyzed using error analysis and explainable AI methods, including LIME [32], to better understand the reasons behind the model's decisions, with a focus on the role of lexical cues and broader context in stress detection. Model uncertainty and calibration were investigated to evaluate the reliability of the predicted confidence [33]–[35]. It is important to note, however, that these models are intended to analyze patterns associated with stress in online text rather than serve as clinical diagnostic tools. The following research questions were considered in this work:

RQ1: How do pretrained and fine-tuned NLP models compare with classical feature-based approaches for stress detection in Reddit text?

RQ2: How do classifier-head-only training, full fine-tuning, and parameter-efficient LoRA compare in predictive performance and parameter efficiency across pretrained transformer models?

RQ3: Can error analysis and explainability methods reveal systematic patterns underlying model errors and the factors influencing model predictions?

RQ4: How well calibrated are the best-performing fully fine-tuned and parameter-efficient transformer models, and how does confidence-based selective prediction affect the tradeoff between predictive performance and coverage?

This paper makes the following contributions:

- Comparing feature-based methods, fixed pretrained transformer embeddings, training only the classifier head, full fine-tuning, and LoRA on the same stress classification task across the same Dreaddit corpus.
- Comparing predictive performance and parameter efficiency of MiniLM, MPNet, and DistilBERT using three random seeds, and conducting a separate ten-seed comparison of full fine-tuning and LoRA-large using matched effective batch sizes and validation-based hyperparameter selection.
- Looking at reliability in addition to overall prediction performance using error analysis, lexical analysis, LIME explanations, confidence analysis, probability calibration, and selective prediction.

The remainder of this paper is organized as follows. Section II reviews related work, Sections III and IV describe the methods and experimental setup, Sections V and VI present and discuss the results, and the remaining sections address conclusions, limitations, ethics, reproducibility, and AI use.

II. Related Work

Computational approaches to mental health and psychological states on social media have evolved from identifying linguistic features in social media to automatically detecting content or individuals that are indicative of specific psychological states. Yates et al. investigated self-harm risk and depression in online community forums. They demonstrated the ability to detect high-risk content and individuals who self-report depression using linguistic analysis [36]. Cohan et al. created the SMHD dataset to study mental health conditions on Reddit. The dataset consists of Reddit users with self-reported mental health diagnoses for multiple mental disorders and matched control users. They performed both classification and linguistic analysis on the dataset [37]. With regard to stress, Guntuku et al. explored linguistic indicators in social media relating to high psychological stress. They found that high stress was associated

with content that focused more on the self, as well as content discussing exhaustion, loss of control, and physical pain [13]. Dreaddit was introduced by Turcan and McKeown, who tested multiple stress classification baselines, such as n-grams, logistic regression, and BERT-base. Their best performing model was BERT-base with an F1-score of 0.8065. A domain-specific logistic-regression model using Word2Vec also performed well with an F1-score of 0.7980 [19]. These studies provide evidence that linguistic indicators of stress and mental health can exist on social media. Furthermore, they suggest that these indicators can vary based on the task, platform, and annotation method.

Recent work has also seen a shift toward using pretrained contextual representations in mental health text classification. Pretrained models can represent context in text, which can then be leveraged for downstream tasks such as classification. Domain-specific pretrained language models, MentalBERT and MentalRoBERTa, were introduced by Ji et al. These models were pretrained on mental health text and were found to improve performance across multiple mental-disorder detection benchmarks [38]. Lokala et al. also explored a knowledge-assisted multitask RoBERTa model on Reddit content to detect mental health indicators in cardiovascular-disease-related content [39]. This approach also leverages pretrained contextual representations along with domain- and task-specific information. These methods suggest the viability of using pretrained language models for mental health classification tasks. This paper builds upon this work to explore multiple degrees of model adaptation on the same stress-classification task. These methods include fixed pretrained transformer representations, training only the classification head, full fine-tuning, and parameter-efficient fine-tuning.

Efficiency, reliability, and interpretability, in addition to predictive performance, have also received substantial attention in recent NLP and machine learning research, particularly with regard to the adaptation of large pretrained models. Ribeiro et al. proposed LIME, which provides local interpretable approximations of individual model predictions in a model-agnostic manner [32]. Reliability can also be characterized through calibration. Guo et al. found that modern neural networks can produce miscalibrated confidence estimates and showed that temperature scaling can serve as an effective post-hoc calibration technique [33]. Miscalibration has also been identified in pretrained language models. Ahuja et al. examined calibration in massively multilingual language models and found that methods including temperature scaling could reduce calibration error in zero-shot settings [40]. Geifman and El-Yaniv explored selective classification, which allows a model to abstain from uncertain predictions at the cost of reduced coverage [35]. There has also been interest in parameter-efficient adaptation techniques that minimize the number of model parameters that need to be adapted for a particular downstream task. Houlsby et al. introduced adapters that approached the performance of full fine-tuning while adding only a small fraction of trainable parameters for each task [41]. Hu et al. subsequently introduced LoRA, which adapts a model to a task through trainable low-rank updates while keeping the pretrained model weights frozen [25].

These efforts highlight the relevance of social media for mental-health- and stress-related language, the utility of pretrained transformers for text classification, and the need to consider interpretability, uncertainty, and efficiency in addition to predictive performance. However, these questions are typically studied across different tasks and experimental conditions. In this study, they are explored together in a stress-classification task on Dreaddit. The performance of models using classical feature representations and fixed pretrained transformer representations is compared with models adapted using only the classification head, full fine-tuning, and LoRA. These models are then examined through error analysis and LIME, while the reliability of their predictions is explored through probability calibration and confidence-based selective prediction. This provides a common experimental setting for examining not only effective modeling approaches for stress detection on Reddit, but also the efficiency of model adaptation and what model errors and confidence estimates reveal about model behavior.

III. Methods

A. Modeling Framework

This paper explores stress classification using two modeling frameworks: feature-based classification and end-to-end transformer classification on the same binary stress-classification task. In feature-based classification, text is encoded into fixed numerical representations and then passed to a separate machine-learning classifier that determines whether the text segment indicates stress. In end-to-end classification, a pretrained transformer is incorporated directly into the classification architecture, and task-specific learning occurs within the transformer model rather than through a downstream classifier. By using these two frameworks on the same classification task, the impact of increasingly task-specific model adaptation can be evaluated. Figure 2 provides an overview of the experiments conducted.

For the feature-based classification framework, text was encoded using Word2Vec [26] as well as the pretrained transformer models MiniLM [29], MPNet [30], and DistilBERT [31]. Each representation was then classified using both XGBoost [28] and Random Forest [27] classifiers. For the end-to-end classification framework, the same three pretrained transformers were adapted using classifier-head-only training, full fine-tuning, and parameter-efficient fine-tuning via LoRA [25]. Using the same transformer families in both frameworks enables a more direct comparison between fixed pretrained representations and task-specific model adaptation and helps to isolate the impact of the training strategy rather than comparing entirely different model families.

B. Feature-Based Classification

The feature-based experiments treated embedding extraction and classification as distinct stages, allowing the same downstream classifiers to be evaluated across multiple fixed text representations.

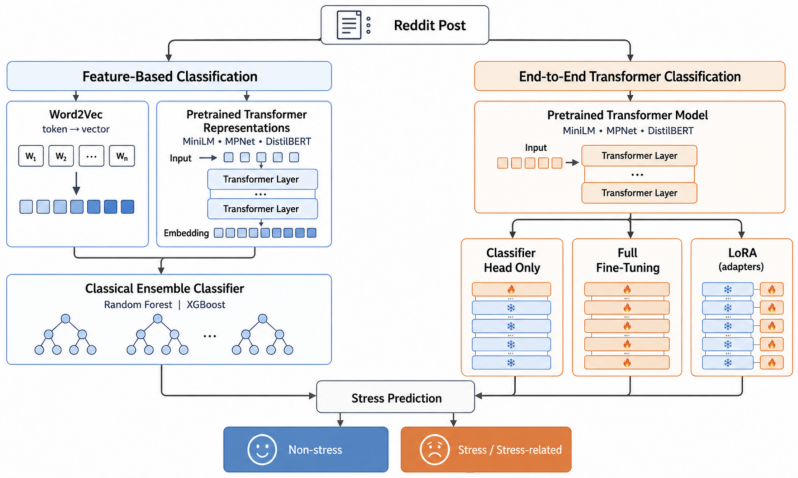


Fig. 2: Overview of the two modeling frameworks evaluated for stress classification. The feature-based approach uses fixed text representations with separate classical classifiers, while the end-to-end approach adapts pretrained transformer models directly for classification using different training strategies. Blue modules indicate frozen parameters and orange modules indicate trainable parameters or low-rank updates in the end-to-end training strategies.

1) Word2Vec Representations: Word2Vec is a representation-learning method that maps individual words to dense numerical vectors learned from surrounding linguistic contexts. Consequently, words occurring in similar linguistic contexts tend to develop similar representations [26]. In this study, the skip-gram formulation of Word2Vec was used, in which the model learns word representations by predicting surrounding context words from a given target word.

Word2Vec was trained on the training text used for the feature-based experiments, where each eligible token received an embedding learned from the Dreaddit training corpus. The classifiers, however, required one fixed-dimensional vector per text segment, so the word embeddings within each segment were combined into a single segment representation. Formally, the segment representation was computed by averaging the embeddings of all in-vocabulary words:

$$\mathbf{x}_s = \frac{1}{n}\sum_{i=1}^{n}\mathbf{v}(w_i), \tag{1}$$

where $\mathbf{x}_s$ denotes the fixed-dimensional vector representation of text segment s, $\mathbf{v}(w_i)$ denotes the Word2Vec embedding of the i-th in-vocabulary word in that segment, and n is the total number of in-vocabulary words included in the average. These fixed segment-level representations were then used as input features for the Random Forest and XGBoost classifiers described in Section III-B3.

2) Pretrained Transformer Representations: Besides Word2Vec, contextual embeddings from pretrained transformer models were also explored for every text. While Word2Vec assigns one learned embedding to each word, embeddings from transformer models utilize contextual information when embedding text. For this feature approach, pretrained transformer models were used as fixed feature extractors. During training of the classifiers, the parameters within the pretrained models remained unchanged. Instead, each text was embedded into a fixed-dimensional vector that was then passed into the same XGBoost and Random Forest classifiers used for the other feature types. Three pretrained transformer models were explored as embedding models: MiniLM [29], MPNet [30], and DistilBERT [31]. MiniLM and MPNet use pretrained SentenceTransformer implementations to create a single embedding for each text at the segment level. DistilBERT, however, creates contextual embeddings at the token level. For each text, the hidden-state vectors of all non-padding tokens in the segment are averaged into a single fixed-dimensional embedding. The embeddings from MiniLM, MPNet, and DistilBERT were then held constant during classifier training and were again assessed with both the XGBoost and Random Forest classifiers, as with the embeddings from Word2Vec.

3) Classical Ensemble Classifiers: The fixed text representations generated in the preceding experiments were evaluated using two classical ensemble classifiers: Random Forest [27] and XGBoost [28]. Random Forest combines the predictions of multiple decision trees trained using different samples and subsets of the available features, with the aggregated predictions reducing the reliance on any individual tree. XGBoost instead uses a gradient-boosting approach in which decision trees are added sequentially to improve the predictions of the existing ensemble while optimizing a regularized objective. In this study, both classifiers were applied to the Word2Vec, MiniLM, MPNet, and DistilBERT representations, allowing the classification performance of the different fixed representations to be compared using the same downstream classification

methods.

C. End-to-End Transformer Classification

1) Pretrained Transformer Backbones: For end-to-end transformer classification, MiniLM [29], MPNet [30], and DistilBERT [31] were used as pretrained transformer backbones. Unlike the feature-based experiments, in which these models were used to generate fixed representations for separate downstream classifiers, the transformer models were incorporated directly into the classification architecture. A classification head was added to each pretrained backbone to produce predictions for the binary non-stress versus stress/stress-related classification task. The extent to which the pretrained backbone was allowed to adapt to the task varied across classifier-head-only training, full fine-tuning, and LoRA.

2) Classifier-Head-Only Training: In classifier-head-only training, the parameters in the pretrained transformer backbone remain frozen while the classification head is trained. Since the transformer backbone parameters are frozen, the pretrained contextual representations cannot be adapted to the Dreaddit stress-classification task; however, a classification decision boundary between non-stress and stress/stress-related can be learned. Hence, this method only offers a limited degree of task-specific adaptation and can be used as a reference point to compare the additional performance achieved by adapting the parameters in the transformer backbone via full fine-tuning or LoRA.

3) Full Fine-Tuning: Under full fine-tuning, unlike classifier-head-only training, both the parameters in the transformer backbone and the classification head are trained. Since all parameters in the model are trained, the pretrained contextual representations can be adapted to the Dreaddit stress-classification task instead of only a new decision boundary being learned. This method thus offers the most comprehensive degree of task-specific adaptation among the standard transformer-training methods used in this paper and can be used as a reference point to assess the degree to which adapting the entire pretrained model results in improved performance compared to classifier-head-only training. However, the increased flexibility offered by full fine-tuning requires training all parameters, and thus comparison can be made to parameter-efficient methods such as LoRA that attempt to maintain task-specific adaptation while adapting a significantly smaller number of parameters.

4) Parameter-Efficient Fine-Tuning with LoRA: LoRA provides task-specific adaptation without updating the entire pretrained parameter set. Rather than modifying all original transformer weights, it introduces small trainable low-rank matrices into selected components of the network. Following Hu et al. [25], a pretrained weight matrix W is kept frozen while the task-specific update is represented by two lower-rank matrices:

$$W' = W + \Delta W, \qquad \Delta W = \frac{\alpha}{r} BA, \tag{2}$$

where r is the low-rank dimension and α controls the scale of the learned update.

For LoRA, trainable low-rank updates were applied to the query and value projections in attention, following the original LoRA approach [25], and left the classification head fully trainable for MiniLM, MPNet, and DistilBERT. No other target-module combinations were evaluated. Different low-rank capacities were evaluated, with larger ranks allowing greater capacity for task-specific adaptation while also increasing the number of trainable parameters. Evaluating multiple LoRA capacities therefore allows the tradeoff between parameter efficiency, task-specific adaptation, and predictive performance to be examined. These LoRA configurations were applied across the same three transformer backbones and compared with full fine-tuning to determine whether comparable predictive performance could be achieved while updating substantially fewer parameters.

D. Error and Explainability Analysis

Since the fully fine-tuned MPNet model was among the strongest-performing configurations and provided the full-parameter baseline for comparison with LoRA, detailed error analysis was conducted to gain insight into the nature of the errors, specifically the distinction between false positives and false negatives. The model's test predictions were split into the four prediction categories: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). First, basic text features such as number of characters, words, sentences, Flesch–Kincaid grade level, sentiment, and prediction confidence were compared between the four prediction categories. Text features were computed directly from the text in each prediction. Readability and sentiment were pulled from the corresponding fields in the Dreaddit dataset [19]. Prediction confidence was obtained from the model's predicted probabilities.

In addition to the general error analysis, Local Interpretable Model-Agnostic Explanations (LIME) were employed to understand the features influencing the model's individual predictions for the fully fine-tuned MPNet model [32]. A reproducible set of 10 false-positive, 10 false-negative, 5 true-positive, and 5 true-negative predictions was used to generate explanations. For each of the selected segments, LIME created 1,000 perturbed samples and kept the 15 features that had the greatest influence on the prediction. Positive feature weights indicated greater influence toward the stress/stress-related class, and negative weights indicated greater influence toward the non-stress class. The explanations were then analyzed individually and for each of the four groups of predictions to find lexical features that commonly influenced both correct and incorrect model predictions. Since LIME provides local approximations of model behavior, the explanations were used to infer the influence of the features rather than as actual expressions of the model's reasoning.

E. Uncertainty Calibration and Selective Prediction

Performance on a classification task does not necessarily mean that the model's predicted probabilities are well calibrated, such that they reflect the likelihood that the prediction is correct.

For stress classification, this can be especially important, as a high-confidence incorrect prediction can be more problematic than an incorrect prediction made with apparent uncertainty. In this study, only predictive uncertainty as reflected by model confidence was examined on the held-out Dreaddit test data. Uncertainty due to distribution shift, label ambiguity, or model parameters was not individually quantified.

Calibration of both the fully fine-tuned MPNet and the MPNet LoRA-large models was assessed, with the LoRA configuration selected based on validation F1-score. Calibration was assessed via Brier scores [34], reliability diagrams, and expected calibration error (ECE, using ten confidence bins). Temperature scaling was fit on the validation logits and then applied to the test logits without additional fitting [33].

Confidence-based selective prediction was also assessed using confidence thresholds between 0.50 and 0.90. All predictions below a given threshold were treated as abstentions. Coverage was recorded, while accuracy and G-mean were calculated for the remaining predictions [35].

IV. Experiments

A. Dataset and Data Preparation

1) Dataset Description: The dataset used for this paper was the Dreaddit dataset, a multi-domain Reddit dataset introduced by Turcan and McKeown for computational stress analysis [19]. The broader corpus contains approximately 190,000 Reddit posts across five categories of Reddit communities. Approximately 3,500 text segments from 3,000 posts were manually labeled for stress, with the labeled dataset being more equally split across the five domains [19]. This study uses the released labeled train/test split, rather than the entire 190,000-post corpus, which contains 2,838 training segments and 715 testing segments, with binary labels corresponding to non-stress and stress/stress-related text.

2) Data Quality and Exploratory Analysis: Before conducting the linguistic and predictive analyses, the dataset was evaluated for several potential quality issues, including missing values, duplicate observations, and overlap between the official training and test sets. No missing text or label values were identified. There were no fully identical rows in the training set, but 18 rows repeated text found in other training rows. Those 18 redundant rows were removed while retaining one copy of each repeated text. Three additional training rows matched text in the official test set and were also removed, leaving the test set unchanged. In total, 21 rows were deleted, and 2,817 rows remained before the training-validation split. As shown in Figure 3a, the original released training and testing splits were also relatively balanced between non-stress and stress/stress-related examples. Following these quality checks, several basic structural characteristics of the text were examined, including word count, character count, sentence count, and average word length. Stress segments averaged approximately 89.02 words compared with 82.02 words for non-stress segments, while they also contained slightly more sentences on average. In contrast, the average word length was slightly lower for stress segments than for non-stress segments. As illustrated in Figure 3b, however, the segment-length distributions exhibited substantial overlap between classes, suggesting that these relatively modest structural discrepancies alone do not clearly distinguish stress from non-stress text.

B. Data Partitioning and Evaluation

All experiments were performed on the cleaned Dreaddit dataset. The original training set was preprocessed to remove duplicate training texts and training examples containing a text from the official test set. This resulted in 2,817 cleaned examples. The remaining examples were split using stratification into 2,394 training examples and 423 validation examples (85% training, 15% validation). There were 201 non-stress examples in the validation set and 222 stress/stress-related examples. The official test set for Dreaddit was preserved as-is with 715 examples, 346 non-stress and 369 stress/stress-related examples [19].

The initial experiments tested classifier-head-only, full fine-tuning, feature-based, and LoRA approaches with seeds 42, 123, and 2026. A subsequent ten-seed experiment directly compared full fine-tuning and LoRA-large using matched effective batch sizes and validation-based hyperparameter selection. The training, validation, and official test sets remained the same between runs; seeds varied stochastic model initialization and training operations where applicable, not the data split. Models, hyperparameters, and checkpoints were chosen based on validation performance. The official test set was used only for evaluation, not model training, hyperparameter selection, checkpoint selection, or temperature fitting.

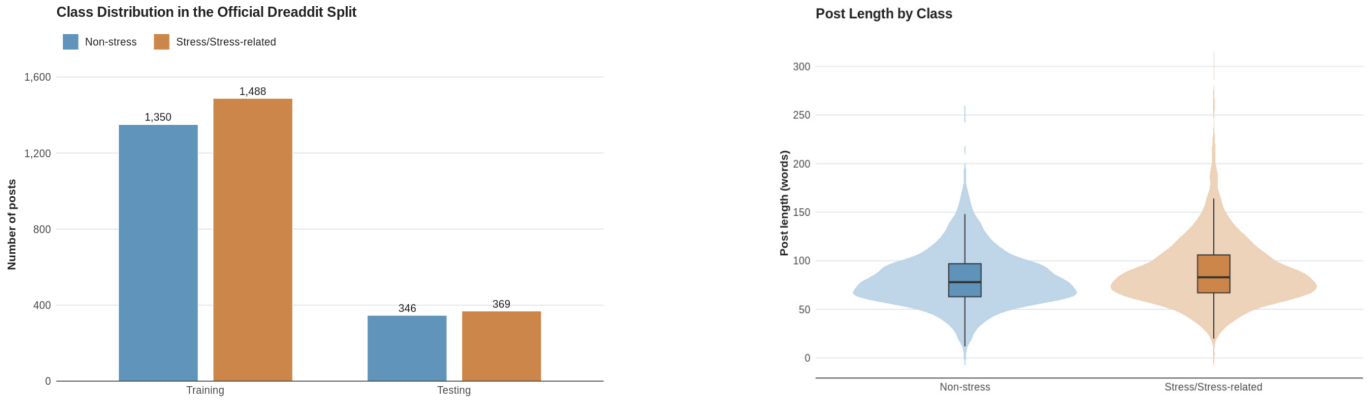
C. Feature-Based Experimental Configuration

The Word2Vec implementation used word tokenization with contraction retention and lower-casing. Skip-gram Word2Vec was fit only on the training set using a 5-token context size, minimum word frequency of 2, 100-dimensional embeddings, and the run-specific random seed. The fixed segment embeddings used in Section III-B1 are the mean of the word embeddings in each segment.

Hyperparameters for XGBoost and Random Forest were tuned on the validation set. For Random Forest, tree count was tested at 100 and 200, maximum tree depth was tested at 10, 20, and unrestricted depth, and class weighting was tested with and without balanced class weighting. For XGBoost, tree count was tested at 100 and 200, maximum tree depth was tested at 3 and 5, learning rate was tested at 0.05 and 0.10, and positive class weight was tested at 0.6 and 1.0. Models were ranked first by validation F1-score, then by recall and ROC-AUC. The best-performing Word2Vec model on the validation set used XGBoost with 200 trees, a maximum depth of 5, a learning rate of 0.10, and a positive class weight of 1.0.

The pretrained-transformer experiments used three checkpoints:

- `sentence-transformers/all-MiniLM-L6-v2`
- `sentence-transformers/all-mpnet-base-v2`
- `distilbert-base-uncased`



(a) (b)

Fig. 3: Overview of the Dreaddit dataset. (a) Class distribution in the official training and testing splits. (b) Distribution of segment length for non-stress and stress/stress-related segments in the cleaned training set.

These checkpoints were used to create MiniLM, MPNet, and DistilBERT encodings, respectively. MiniLM created segment encodings of 384 dimensions, while MPNet and DistilBERT created 768-dimensional encodings. MiniLM and MPNet encodings were created via the SentenceTransformer implementation with batch sizes of 32 and 16, respectively. DistilBERT encodings were created by truncating the text to a maximum length of 256 tokens and then mean-pooling the contextual encodings of all non-padding tokens to create a single segment encoding.

In order to facilitate comparison between the downstream classifiers for all three of the fixed encodings based on transformers, the same preconfigured XGBoost and Random Forest classifiers were used for each encoding. The Random Forest used 200 trees with balanced class weighting and no restriction on tree depth. The XGBoost used 200 trees with a learning rate of 0.05 and a maximum depth of three. All models were trained on the standard training set and tested on the official test set.

D. End-to-End Transformer Training Configuration

The same MiniLM, MPNet, and DistilBERT models that were tested on fixed text encodings were also tested with end-to-end training. Text pieces were tokenized according to their respective checkpoints, truncated to a maximum sequence length of 256 tokens, and dynamically padded within minibatches.

Classifier-head-only training kept the pretrained transformer frozen but trained the classifier heads with a learning rate of 2×10^{-4}, a batch size of 16 during training, and no gradient accumulation. Full fine-tuning trained all model parameters with a learning rate of 2×10^{-5}, a batch size of eight during training, and two steps of gradient accumulation. Both methods were trained for three epochs with a batch size of 16 during evaluation, a weight decay of 0.01, and a warmup ratio of 0.10. AdamW was used for optimization. After each epoch, the model was evaluated on the validation set and the checkpoint with the highest validation F1-score was used for final evaluation.

Three different low-rank configurations were tested on the three transformer models using LoRA. For LoRA-small, $r = 4$ and $\alpha = 8$ were used. For LoRA-medium, $r = 8$ and $\alpha = 16$ were used. For LoRA-large, $r = 16$ and $\alpha = 32$ were used. In all three cases, a dropout rate of 0.10 was used. For each model, the low-rank updates occurred in the query and value projections, but the classification layers remained trainable. All LoRA configurations were trained for three epochs with a learning rate of 2×10^{-4}, a batch size of eight for training and 16 for evaluation, a weight decay of 0.01, and a warmup ratio of 0.10. The checkpoint with the highest F1-score on the validation set was saved, as was done in the other transformer experiments.

Since full fine-tuning and LoRA used different training settings in the original experiments, another experiment was run using the same effective batch size (eight) for both approaches. For full fine-tuning, learning rates of 10^{-5}, 2×10^{-5}, and 5×10^{-5} were tried, and for LoRA-large, learning rates of 10^{-4}, 2×10^{-4}, and 5×10^{-4} were tried. For each backbone and training method, the learning rate was selected using validation F1-score from seed 42. Training was performed for a maximum of five epochs and the best checkpoint was picked using validation F1-score. The chosen settings were then tried with ten paired random seeds. Exact paired sign-flip permutation tests with Holm correction for six comparisons were run on the ROC-AUC and F1-score differences. MPNet full fine-tuning selected the highest tested learning rate (5×10^{-5}), with validation F1 rising from 0.832 to 0.847 to 0.854 across the grid; the optimum may lie above the tested range. DistilBERT and MiniLM full fine-tuning and DistilBERT LoRA-large selected the lowest tested learning rate, so their optima may also lie outside the tested grids.

E. Evaluation Metrics

Performance was assessed using accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve (ROC-AUC). For precision, recall, and F1-score, stress/stress-related text was used as the positive label. The

counts of true positives, true negatives, false positives, and false negatives were also stored to facilitate later investigation of systematic errors.

As part of investigating the contrast between full fine-tuning and parameter-efficient methods, the number of total and trainable parameters within the transformer models was also stored. For each LoRA experiment, the percentage of parameters that were trainable and the reduction in trainable parameters compared to normal full fine-tuning were also recorded alongside performance metrics.

Calibration was assessed using reliability diagrams, expected calibration error (ECE), and Brier score for the uncertainty analysis. Temperature scaling was fitted only on the validation logits and applied to the test logits without further fitting. Model confidence was taken to be the maximum of the predicted probabilities for each class in selective prediction based on confidence. Confidence thresholds of 0.50, 0.60, 0.70, 0.80, and 0.90 were evaluated, and performance and coverage were measured only for predictions having confidence greater than or equal to the set confidence threshold.

F. Implementation Environment

The experiments were written in Python and run in Google Colab on an NVIDIA A100-SXM4-40GB GPU. For the transformer and LoRA experiments, the following software versions were used: PyTorch 2.11.0 with CUDA 13.0 (cu130), Transformers 5.16.1, Sentence-Transformers 6.0.1, Datasets 4.8.4, PEFT 0.20.0, Accelerate 1.14.0, and PyArrow 23.0.1.

Other packages used in the larger experimental pipeline included scikit-learn for data splitting, classical classification and evaluation, XGBoost for gradient-boosted classification, Gensim for training the Word2Vec model, LIME for local explainability, and NumPy, pandas, Matplotlib, and plotnine for data manipulation and data visualisation. The archived requirements-notebook.txt lists scikit-learn 1.6.1, XGBoost 3.4.1, Gensim 4.4.0, and LIME 0.2.0.1. These describe the archived notebook environment and should not be assumed to document every earlier run. Mixed-precision training was used for the transformer experiments where CUDA was available. The three-seed empirical-reporting reruns were conducted on an NVIDIA A100-SXM4-40GB GPU with an Intel Xeon CPU at 2.20 GHz, 12 logical CPUs, and 83.47 GB of system RAM. Across the transformer configurations, mean training time ranged from 10.12 to 61.51 seconds, mean evaluation time ranged from 0.59 to 1.31 seconds, and mean peak GPU memory ranged from 0.56 to 3.60 GB.

V. Results

A. Comparison of Feature-Based and End-to-End Classification

The first experiments explored the impact of representational complexity on performance in stress classification. In the embedding-based experiments, Word2Vec embeddings trained on the Dreaddit training data were evaluated alongside fixed contextual embeddings generated using MiniLM, MPNet, and DistilBERT. For each of the fixed transformer embeddings, both XGBoost and Random Forest were used. This enabled the impact of the underlying representation to be examined while retaining a traditional downstream classifier in the system.

After hyperparameter tuning on the validation set, the best-performing Word2Vec configuration was evaluated on the official test set using XGBoost. Across the three seeds, this model achieved an average accuracy of 0.699 ± 0.021, ROC-AUC of 0.773 ± 0.013, and F1-score of 0.731 ± 0.018.

The next analysis examined whether replacing the Word2Vec representations with pretrained transformer embeddings improved performance. For MiniLM embeddings, XGBoost achieved an average F1-score of 0.769 ± 0.000 compared with 0.753 ± 0.017 for Random Forest. For MPNet embeddings, Random Forest and XGBoost achieved average F1-scores of 0.767 ± 0.002 and 0.760 ± 0.000, respectively. DistilBERT embeddings yielded the strongest results among the fixed representations, with XGBoost achieving an average accuracy of 0.769 ± 0.000 and F1-score of 0.787 ± 0.000. The three-seed test-set results for the Word2Vec baseline and six fixed-transformer configurations are summarized in Table I.

This indicates that the contextual embeddings from pretrained transformers were able to represent features useful for distinguishing stress and non-stress text, without the actual transformers themselves being fine-tuned on the Dreaddit dataset. DistilBERT improved on Word2Vec (F1 0.787 versus 0.731). For context, Turcan and McKeown's published BERT-base F1-score of 0.8065 lies below the matched MPNet LoRA-large result (0.8473) but above the best fixed-embedding result (0.7871); the experimental protocols differed [19]. The most effective downstream classifier also varied between representations: XGBoost yielded the best results for both DistilBERT and MiniLM, whereas slightly better average F1-score was achieved with Random Forest for MPNet. This suggests that performance in the feature-based approach was affected by both the representation and the downstream classifier used to interpret it.

The fixed transformer representations showed considerable improvement relative to the Word2Vec representations but were produced by transformer models that had not been adapted to the Dreaddit stress-classification task. This motivated evaluating whether additional value could be obtained by allowing the transformer backbone itself to adapt to the task. Thus, the three transformer backbones were tested both with classifier-head-only training and full fine-tuning.

The results of fine-tuning for the specific task differ between the three transformer models. For MiniLM, training only the classifier head resulted in an average accuracy of 0.518 ± 0.001 and F1-score of 0.682 ± 0.000. MiniLM showed high recall for stress in the seed-42 experiment with 714 out of 715 samples being classified as stress. All 369 stress samples were detected, but just one non-stress sample was correctly identified. This high recall for stress largely reflects the model's tendency to predict a single class; in this run, it effectively failed to distinguish stress from non-stress text. Full fine-tuning resulted in an average accuracy of 0.779 ± 0.015, F1-score of 0.805 ± 0.006, and ROC-AUC of 0.867 ± 0.010.

For MPNet, training only the classifier head resulted in an

TABLE I: Three-seed test-set performance of feature-based stress-classification methods

Representation	Downstream classifier	Accuracy	G-mean	F1-score	ROC-AUC
Word2Vec	XGBoost	69.93 ± 2.07	68.99 ± 2.13	73.07 ± 1.81	77.30 ± 1.29
MiniLM	XGBoost	75.24 ± 0.00	74.93 ± 0.00	76.92 ± 0.00	82.16 ± 0.00
MiniLM	Random Forest	73.38 ± 1.66	73.00 ± 1.62	75.29 ± 1.67	81.55 ± 0.46
MPNet	XGBoost	74.69 ± 0.00	74.51 ± 0.00	76.03 ± 0.00	83.70 ± 0.00
MPNet	Random Forest	75.48 ± 0.49	75.33 ± 0.58	76.71 ± 0.21	83.32 ± 0.42
DistilBERT	XGBoost	**76.92 ± 0.00**	**76.50 ± 0.00**	**78.71 ± 0.00**	**85.60 ± 0.00**
DistilBERT	Random Forest	75.52 ± 0.61	75.17 ± 0.53	77.26 ± 0.74	84.24 ± 0.32

Note. Values are mean ± standard deviation across seeds 42, 123, and 2026 and are reported on a 0–100 scale. Fixed transformer embeddings were deterministic and XGBoost did not use stochastic subsampling, so their three identical fits (SD 0.00) do not provide evidence of variation across randomized runs. Stress/stress-related text was treated as the positive class. G-mean is the geometric mean of sensitivity and specificity. Bold indicates the highest mean value in each performance column. Published Dreaddit F1 benchmarks from Turcan and McKeown [19] are majority (68.08), n-grams (74.41), n-grams with features (77.00), logistic regression with domain-specific Word2Vec and features (79.80), and BERT-base (80.65). These are external reference results obtained under a different experimental protocol.

average accuracy of 0.759 ± 0.004 and F1-score of 0.781 ± 0.003. Full fine-tuning resulted in an average accuracy of 0.812 ± 0.013 and F1-score of 0.830 ± 0.009. The average ROC-AUC increased from 0.854 ± 0.001 to 0.893 ± 0.013.

For DistilBERT, training only the classifier head resulted in an average F1-score of 0.784 ± 0.002. Full fine-tuning increased the average F1-score to 0.814 ± 0.011 and the average ROC-AUC increased from 0.843 ± 0.005 to 0.880 ± 0.004. Average accuracy also increased from 0.759 ± 0.006 to 0.802 ± 0.009. A comparison of three-seed performance between full fine-tuning and classifier-head-only training on the test set can be found in Table II.

For the classifier-head-only, feature-based, and fully fine-tuned models used in this phase, the fully fine-tuned MPNet model achieved the strongest average performance with a ROC-AUC of 0.893 ± 0.013, an F1-score of 0.830 ± 0.009, and an accuracy of 0.812 ± 0.013. This highlights the advantage of letting pretrained transformers adapt their representations to the stress classification task. Full fine-tuning also involved updating the entire transformer, so the next experiment looks at whether similar performance can be achieved with a very small number of trainable parameters.

B. Parameter-Efficient Fine-Tuning

Fully fine-tuned models required all parameters in each transformer model to remain trainable. For the fully fine-tuned models, MiniLM had 22.7 million trainable parameters, DistilBERT had 67.0 million trainable parameters, and MPNet had 109.5 million trainable parameters. Thus, a substantial number of trainable parameters was required to achieve the best model performance. To address this, experiments were conducted to determine if LoRA could maintain the advantage of task-specific adaptation but with only a fraction of the trainable parameters required by full fine-tuning [25].

For each of the three transformer models, LoRA only made a small fraction of the parameters in the model trainable during fine-tuning, but in many cases achieved performance comparable to full fine-tuning. This was particularly true for MPNet. Full fine-tuning achieved an average F1-score of 0.830 ± 0.009. LoRA-small, LoRA-medium, and LoRA-large achieved average F1-scores of 0.836 ± 0.005, 0.837 ± 0.005, and 0.839 ± 0.005, respectively. The performance between these three models was very similar, but each had about 0.67%, 0.80%, and 1.07% of their parameters in the model trainable. This represents reductions from full fine-tuning of approximately 99.32%, 99.19%, and 98.92%, respectively.

Across the three-seed averages for MPNet, LoRA-large achieved the highest F1-score and ROC-AUC among the three LoRA configurations. However, LoRA-large had already been chosen as the MPNet configuration for the following reliability tests because it achieved the highest validation F1-score (0.858) in the original seed-42 experiment. It should be noted that this choice was based on validation performance and not test performance.

For DistilBERT, full fine-tuning and the three LoRA configurations achieved very similar average F1-scores. Full fine-tuning achieved an average F1-score of 0.814 ± 0.011, while LoRA-small, LoRA-medium, and LoRA-large achieved 0.813 ± 0.003, 0.812 ± 0.005, and 0.815 ± 0.009, respectively. LoRA-small achieved this performance by making about 1% of the model parameters trainable, representing a reduction of approximately 99.01% in trainable parameters as compared to full fine-tuning. The DistilBERT results also indicate that increasing the rank in the low-rank update does not necessarily correlate with improved performance.

For MiniLM, full fine-tuning achieved the highest average F1-score compared to the three LoRA configurations. Full fine-tuning achieved an average F1-score of 0.805 ± 0.006 compared to 0.782 ± 0.008, 0.795 ± 0.004, and 0.799 ± 0.007 for LoRA-small, LoRA-medium, and LoRA-large, respectively. The performance reduction between the three LoRA configurations and full fine-tuning was relatively small. However, LoRA-large was only about 0.006 F1 points below full fine-tuning while making only about 0.65% of the parameters in the model trainable. For LoRA-small, only about 0.17% of parameters were made trainable in the model. This represents a reduction of approximately 99.83% in trainable parameters as compared to full fine-tuning. In Table III one can find a summary of the number of parameters made trainable and three-seed test performance for full fine-tuning and the three LoRA configurations.

Across the three-seed experiments, MPNet LoRA configurations achieved slightly higher average F1-scores than MPNet full fine-tuning, DistilBERT LoRA and full fine-tuning achieved very similar average F1-scores, and MiniLM full fine-tuning

TABLE II: Three-seed test-set performance of classifier-head-only training and full fine-tuning

Backbone	Training method	Accuracy	G-mean	F1-score	ROC-AUC
MiniLM	Classifier-head-only	51.79 ± 0.08	6.12 ± 1.29	68.17 ± 0.04	67.09 ± 1.96
MiniLM	Full fine-tuning	77.86 ± 1.50	76.61 ± 2.29	80.49 ± 0.55	86.72 ± 1.03
MPNet	Classifier-head-only	75.94 ± 0.37	75.35 ± 0.41	78.08 ± 0.29	85.36 ± 0.08
MPNet	Full fine-tuning	**81.17 ± 1.34**	**80.44 ± 1.63**	**83.04 ± 0.91**	**89.29 ± 1.33**
DistilBERT	Classifier-head-only	75.94 ± 0.56	75.07 ± 1.12	78.39 ± 0.24	84.33 ± 0.48
DistilBERT	Full fine-tuning	80.19 ± 0.93	79.97 ± 0.85	81.36 ± 1.08	88.02 ± 0.42

Note. Values are mean ± standard deviation across seeds 42, 123, and 2026 and are reported on a 0–100 scale. Stress/stress-related text was treated as the positive class. G-mean is the geometric mean of sensitivity and specificity. Bold indicates the highest mean value in each performance column.

TABLE III: Three-seed test-set performance and parameter efficiency of full fine-tuning and LoRA

Backbone	Training method	Accuracy	G-mean	F1-score	ROC-AUC	Trainable parameters	Trainable (%)
MPNet	Full fine-tuning	81.17 ± 1.34	80.44 ± 1.63	83.04 ± 0.91	89.29 ± 1.33	109,488,002	100.00
MPNet	LoRA-small	82.38 ± 0.48	82.07 ± 0.53	83.61 ± 0.45	90.70 ± 0.32	739,586	0.67
MPNet	LoRA-medium	82.61 ± 0.40	82.35 ± 0.42	83.73 ± 0.51	90.76 ± 0.46	887,042	0.80
MPNet	LoRA-large	**82.75 ± 0.49**	**82.47 ± 0.57**	**83.90 ± 0.45**	**91.06 ± 0.31**	1,181,954	1.07
DistilBERT	Full fine-tuning	80.19 ± 0.93	**79.97 ± 0.85**	81.36 ± 1.08	88.02 ± 0.42	66,955,010	100.00
DistilBERT	LoRA-small	79.77 ± 0.32	79.39 ± 0.32	81.29 ± 0.32	87.91 ± 0.03	665,858	0.98
DistilBERT	LoRA-medium	79.86 ± 0.37	79.55 ± 0.30	81.25 ± 0.48	88.04 ± 0.25	739,586	1.09
DistilBERT	LoRA-large	**80.23 ± 0.77**	79.97 ± 0.71	**81.51 ± 0.85**	**88.07 ± 0.36**	887,042	1.31
MiniLM	Full fine-tuning	77.86 ± 1.50	76.61 ± 2.29	**80.49 ± 0.55**	86.72 ± 1.03	22,713,986	100.00
MiniLM	LoRA-small	75.29 ± 2.86	73.71 ± 4.85	78.21 ± 0.78	85.38 ± 1.33	37,634	0.17
MiniLM	LoRA-medium	77.62 ± 0.37	77.10 ± 0.41	79.52 ± 0.44	86.83 ± 0.14	74,498	0.33
MiniLM	LoRA-large	**78.09 ± 0.53**	**77.61 ± 0.44**	79.87 ± 0.71	**87.25 ± 0.21**	148,226	0.65

Note. Values are mean ± standard deviation across seeds 42, 123, and 2026 and are reported on a 0–100 scale. Stress/stress-related text was treated as the positive class. G-mean is the geometric mean of sensitivity and specificity. Trainable (%) uses the PEFT-reported denominator, which counts both the frozen original classification head and its saved trainable copy (MPNet LoRA-large: $1{,}181{,}954/110{,}669{,}956 = 1.068\%$). Bold indicates the highest mean performance value within each backbone. Model configurations were selected using validation-set performance before test-set evaluation. MPNet LoRA-large was used in the subsequent reliability analyses because it achieved the highest validation F1-score among the MPNet LoRA configurations in the original seed-42 experiment.

achieved a slightly higher average F1-score than its LoRA configurations. Additionally, there does not appear to be a consistent monotonic relationship between LoRA rank and performance across backbones, indicating that increasing the capacity of the low-rank update does not necessarily correlate with improved performance on this task. MPNet full fine-tuning and LoRA-large were also used in the reliability analysis described later in the study.

Matched training conditions were used to determine whether the performance differences persisted. Ten paired random seeds were tested to compare full fine-tuning and LoRA-large. For MPNet, LoRA-large attained an F1-score of 0.8473 versus 0.8217 for full fine-tuning and a ROC-AUC of 0.9125 versus 0.8941. The mean F1-score gain across paired runs was 0.0256 (95% bootstrap confidence interval: 0.0192–0.0326), and the mean gain in ROC-AUC was 0.0184 (95% CI: 0.0100–0.0293). Both differences were significant following Holm correction (adjusted $p = 0.0117$). For DistilBERT, the ROC-AUC difference was significant following Holm correction (adjusted $p = 0.0156$), but the F1-score difference was not (adjusted $p = 0.3828$). For MiniLM, neither the ROC-AUC difference (adjusted $p = 0.0645$) nor the F1-score difference (adjusted $p = 0.3828$) was significant following Holm correction.

C. Error Patterns of the Fully Fine-Tuned MPNet Model

The original selected fully fine-tuned MPNet model (from the three-seed design, not the matched seed-42 rerun) was among the best-performing systems and was used in the detailed error analysis. However, overall performance metrics do not provide information on the kind of errors that a system produces, or potential causes for those errors. The confusion matrix showed some imbalance in the types of errors. On the official test set, which consists of 715 examples, MPNet generated 328 true positives, 256 true negatives, 90 false positives, and 41 false negatives. There were roughly 2.20 times more false positives than false negatives.

This indicates that MPNet was more likely to classify a non-stress example as stress/stress-related than to fail to classify a stress example. The official test set is fairly well balanced with 346 non-stress examples and 369 stress/stress-related examples. Thus, this imbalance in error types cannot be readily attributed to a large class imbalance in the test set.

Several features were examined to determine if misclassified examples were different from correctly classified examples. False-positive examples averaged 88.99 words while true negatives averaged 79.98 words. For examples with the same true non-stress label this difference was about nine words. False positives also demonstrated somewhat reduced average confidence in the prediction (0.882) relative to true negatives (0.911). False positives also demonstrated somewhat reduced average sentiment values (0.038) relative to true negatives (0.134).

Confidence values showed a more significant difference for stress examples. False negatives had a mean confidence of about 0.839 compared to 0.927 for true positives. False negatives also had somewhat fewer words per example (approximately 83.32) than true positives (approximately 89.36). Differences in sentence count and Flesch–Kincaid score were not particularly significant. There does not appear to be a simple readability or structural feature that readily explains the observed classifi-

cation differences.

The lower confidence in the errors, especially in false negatives, seems to suggest that model uncertainty also holds information regarding the reliability of the predictions. Before turning to a direct analysis of prediction reliability, first the text in the errors was examined to see if there were any specific textual features that could account for the misclassification of some non-stress instances as stress.

D. Lexical Error Analysis and Model Explainability

Beyond these basic structural characteristics, the training data were further examined for differences in the language used by stress and non-stress segments, using unigrams, bigrams, TF-IDF, and readability. For the unigram and bigram analyses, frequencies were normalized within each class to account for differences in the total amount of text, with unigram frequencies expressed per 1,000 tokens and bigram frequencies per 1,000 bigram positions. The difference between the stress and non-stress frequencies was then calculated for each feature where positive values indicate greater frequency of a given unigram or bigram in stress segments and negative values indicate greater frequency in non-stress segments. For TF-IDF analysis, the text was lowercased, a custom set of common stop words was removed, and unigram and bigram features were extracted from a vocabulary limited to 5,000 terms; mean TF-IDF values were then calculated separately for each class, with the difference between the stress and non-stress means calculated in the same direction. Readability was additionally evaluated using the Flesch–Kincaid grade level. Figure 4 summarizes the resulting lexical and readability differences between the two classes.

The unigram and bigram analyses reveal several differences in the language characteristic of the two classes. As shown in Figure 4a, first-person and self-referential terms—such as “I”, “my”, “me”, and “I’m”—and affect-related terms—such as “feel”—occurred more frequently in stress segments. It is also important to note that the non-stress segments exhibit a different lexical profile, including greater prominence of terms such as “you”, “her”, “she”, and “we”. Similar distinctions appear in the bigram analysis in Figure 4b, where phrases including “feel like,” “don’t know,” “know what,” and “right now” were more frequent in stress segments, while phrases such as “mental health,” “each other,” “years ago,” and “first time” were more characteristic of non-stress text.

The TF-IDF analysis in Figure 4c further reflects these lexical differences. Similar to the findings from the unigrams and bigrams, the positive TF-IDF scores of self-referential words and affect-related language indicate a greater association between such language and the stress class while more impersonal words have the opposite effect. While these lexical analyses revealed visible distinctions between the two classes, Figure 4d shows that readability provided considerably less separation: the Flesch–Kincaid grade level was almost identical between classes, approximately 7.56 for non-stress and 7.67 for stress.

Taken together, these analyses suggest that the two classes differ more clearly in their lexical content than in their basic structural characteristics or readability, motivating the subsequent evaluation of computational models designed to capture these linguistic patterns.

By using normalized unigram and bigram analysis, similarities in language between correctly predicted stress and false positives were found. First-person and self-referential unigrams such as “I’m” and “I’ve” were common in false positives. Additionally, the unigrams “don’t”, “really”, and “feel” were also common. The bigram “feel like” was also very common in false positives. These results are consistent with the exploratory analysis described earlier, which found that first-person and affect-related language was more common in the stress class.

Another way to show this trend is to look more closely at some of the stress-related language. Variants of “can’t” and “cannot” appeared about 3.24 times per 1,000 words in true positives, and only 0.59 times per 1,000 words in true negatives. In false positives, they appeared about 1.62 times per 1,000 words. The word “worried” appeared about 0.25 times per 1,000 words in false positives and about 0.05 times per 1,000 words in true negatives. Furthering these differences, the word “sad” appeared in false positives but not in true negatives. While the frequencies of these individual words are low and should be interpreted with caution, the overall trend does show that some non-stress text contain words similar to those that appear in correctly predicted stress. The normalized frequencies of these selected stress-related terms across the four prediction outcomes are summarized in Figure 5.

In order to investigate the impact of these words on individual predictions, LIME was used to explain individual instances from the reproducible sample described in Section III.D. In these explanations, first-person words were seen multiple times contributing toward both correct and incorrect stress predictions. The word “I” contributed positively toward the stress class in 7 out of 10 false-positive predictions. The word “my” was also contributing positively toward the stress class in 2 out of the 10 false-positive predictions. Affect-related words like “angry”, “sad”, “sick”, “ashamed”, and “feel” also appeared as positive contributors in some of the false-positive predictions. In correctly predicted stress, the words “I”, “me”, “scared”, and “feel” also appeared multiple times as positively contributing toward the stress class.

False-negative explanations also helped clarify that these results should not be taken to indicate that individual words are always sufficient to determine the model’s predictions. For instance, words like “anxiety,” “panic,” “struggling,” and “emotional” could still contribute positively toward the stress class in examples that were ultimately predicted as non-stress. Thus, the final prediction captures the combination of multiple signals as well as the overall representation the model has learned, rather than simply being determined by the presence of certain words. The recurring stress-directed features identified in the sampled false-positive and false-negative LIME explanations are summarized in Figure 6.

The overlap between the lexical patterns identified in the normalized frequency analysis and the features highlighted by LIME suggests that affective and self-referential language

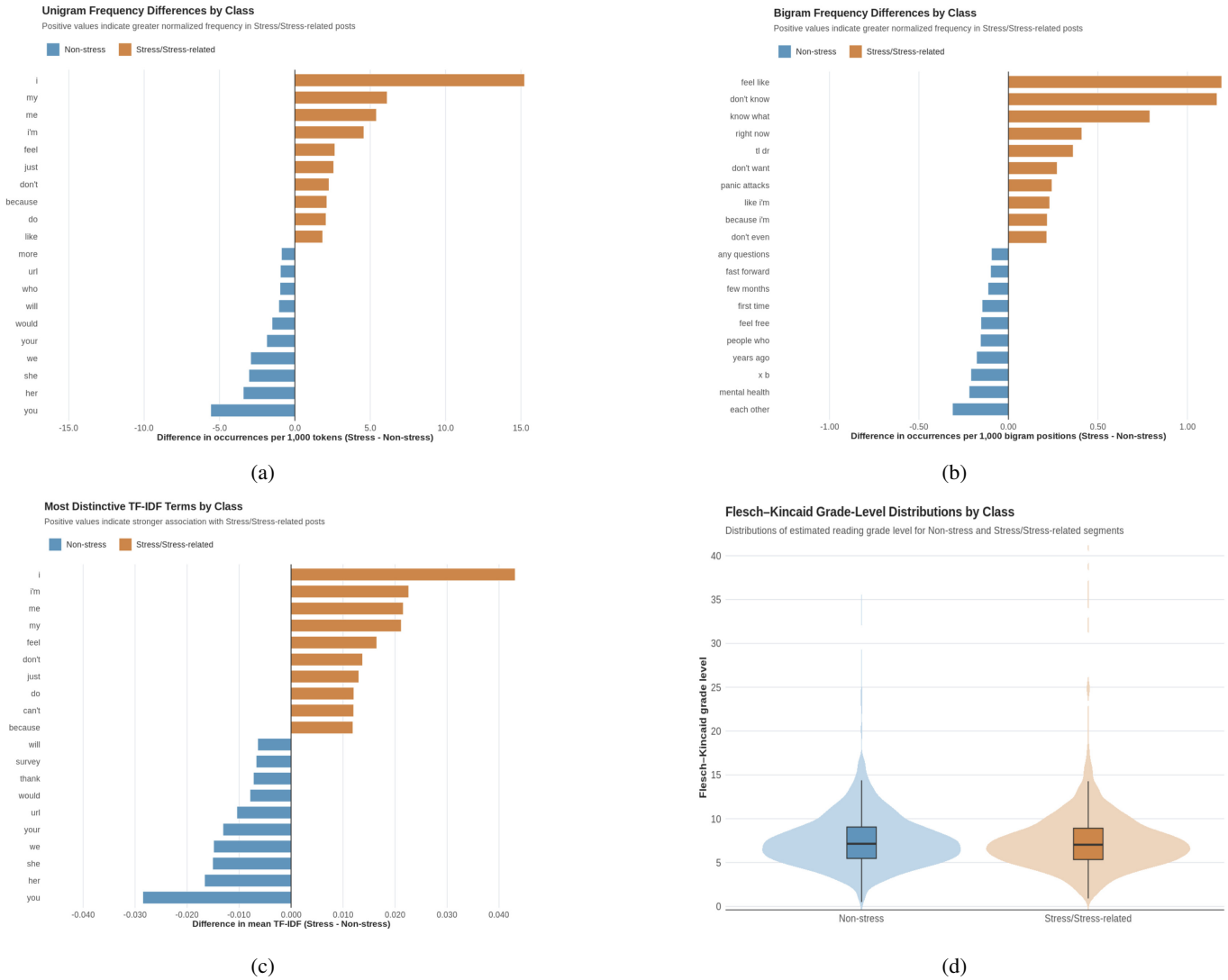


Fig. 4: Exploratory linguistic differences between non-stress and stress/stress-related segments in the cleaned Dreaddit training data. (a) Most distinctive unigrams by normalized frequency difference. (b) Most distinctive bigrams by normalized frequency difference. (c) Difference in mean TF-IDF between stress and non-stress segments. (d) Flesch–Kincaid grade-level distributions by class. Positive values in (a)–(c) indicate greater association with stress/stress-related text, while negative values indicate greater association with non-stress text.

may systematically influence the model's predictions. This is also aligned with the larger potential issue that text classifiers can use superficial or spuriously correlated features in making predictions if those features are correlated with the target label [24]. However, since LIME only provides a local approximation of model behavior and not an actual representation of how the model makes its decisions, these results should be taken as evidence of the influence of multiple signals and not as evidence that the model relies only on lexical signals in making its predictions.

Although incorrect predictions showed lower mean confidence than correct predictions, the model remained relatively confident even when its predictions were wrong. False positives had a mean confidence of 0.882 and false negatives had a mean confidence of 0.839, raising the question of whether the model's predicted probabilities accurately reflected the reliability of its predictions. This motivated a final analysis of probability calibration and whether model uncertainty could be used to identify predictions that warrant more careful evaluation.

E. Probability Calibration and Confidence-Based Selective Prediction

In general, incorrect predictions had lower confidence than correct predictions, but in both MPNet configurations a number of highly confident wrong predictions were observed. For the fully fine-tuned model, correct predictions had an average confidence of 0.920 while incorrect predictions had an average confidence of 0.868. For LoRA-large, correct predictions had an average confidence of 0.928 while incorrect predictions had an average confidence of 0.814. This indicates that confidence

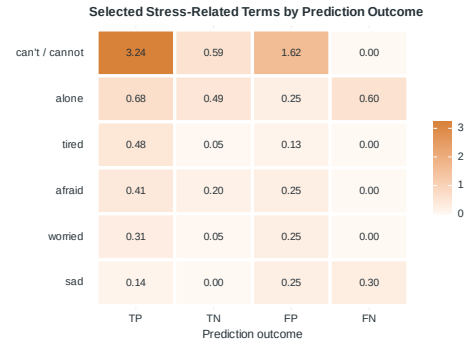


Fig. 5: Normalized frequencies of selected stress-related terms across true-positive, true-negative, false-positive, and false-negative MPNet predictions. Values represent occurrences per 1,000 words within each prediction group.

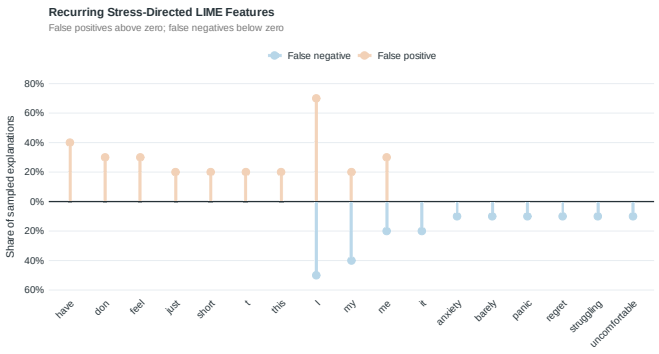


Fig. 6: Recurring stress-directed features in LIME explanations for false-positive and false-negative predictions. Markers indicate the proportion of sampled explanations in which each feature contributed toward the stress/stress-related class.

in the model has some relationship with prediction correctness; however, confidence alone is not sufficient to reliably filter out incorrect predictions.

Temperature scaling improved calibration without changing predicted class labels. In the matched ten-seed follow-up, MPNet LoRA-large mean ECE decreased from 0.1003 to 0.0434 and mean Brier score from 0.1342 to 0.1237; MPNet full fine-tuning mean ECE decreased from 0.1415 to 0.0691 and mean Brier score from 0.1721 to 0.1499. ECE used ten confidence bins. For the original fully fine-tuned model, ECE improved from 0.1010 to 0.0616. The Brier score also improved from 0.1499 to 0.1404. For the original LoRA-large model, ECE improved from 0.0887 to 0.0343 and the Brier score improved from 0.1383 to 0.1287. Accuracy and G-mean remained constant and ROC-AUC was essentially unchanged by temperature scaling. This is because temperature scaling modifies the magnitude of the predicted probabilities but does not alter the ordering of these probabilities or the predicted class labels.

With the temperature-scaled probabilities, confidence-based selective prediction was also tested. With a confidence threshold of 0.80, 84.34% of the test set was retained by the fully fine-

TABLE IV: Calibration before and after temperature scaling in the original selected-model analysis (three-seed design, selected models; not matched ten-seed runs)

Model	Probabilities	Temperature	ECE	Brier score
Full FT	Original	1.0000	0.1010	0.1499
Full FT	Temperature-scaled	1.5341	**0.0616**	**0.1404**
LoRA-L	Original	1.0000	0.0887	0.1383
LoRA-L	Temperature-scaled	1.7399	**0.0343**	**0.1287**

Note. Full FT denotes MPNet full fine-tuning, and LoRA-L denotes MPNet LoRA-large. Lower ECE and Brier scores indicate better calibration. Bold indicates the better result within each model.

TABLE V: Selective-prediction performance using temperature-scaled probabilities in the original selected-model analysis (three-seed design, selected models; not matched ten-seed runs)

Model	Threshold	Coverage (%)	Accuracy (%)	G-mean (%)
Full FT	0.50	100.00	81.68	81.10
Full FT	0.60	96.36	83.02	82.38
Full FT	0.70	93.29	83.21	82.54
Full FT	0.80	84.34	85.90	85.10
Full FT	0.90	0.00	—	—
LoRA-L	0.50	100.00	81.96	81.63
LoRA-L	0.60	90.91	85.23	84.85
LoRA-L	0.70	80.28	87.11	86.55
LoRA-L	0.80	69.79	89.18	88.23
LoRA-L	0.90	46.85	94.63	93.29

Note. Full FT denotes MPNet full fine-tuning, and LoRA-L denotes MPNet LoRA-large. Coverage, accuracy, and G-mean are reported on a 0–100 scale. Accuracy and G-mean were calculated only over predictions retained at or above the specified confidence threshold. No temperature-scaled Full FT predictions reached a confidence of 0.90.

tuned model with 85.90% accuracy on the retained samples as opposed to 81.68% accuracy on all samples. With the same confidence threshold, 69.79% of the test set was retained by LoRA-large with 89.18% accuracy on the retained samples. The initial performance of LoRA-large at a confidence threshold of 0.90 showed 46.85% of test data retained with 94.63% accuracy compared to none retained with full fine-tuning. The follow-up with ten seeds showed that on average LoRA-large retained 47.33% of predictions with 94.74% accuracy compared to 3.52% retained for full fine-tuning. For full fine-tuning, in 8 out of the 10 runs no predictions achieved a calibrated confidence of at least 0.90. The archived ten-seed logits suggest a possible mechanical explanation for the different confidence distributions: the median maximum logit margin per run was 4.59 for full fine-tuning versus 7.26 for LoRA-large, despite similar overall median margins (4.22 versus 4.23). Median fitted temperatures were 2.23 and 1.99, respectively. The different classifier-head learning rates (5×10^{-5} versus 2×10^{-4}) and head activations may contribute: MPNet and MiniLM use tanh, whereas DistilBERT uses ReLU. This is a plausible mechanism, not an established causal result. When full fine-tuning retained predictions at 0.90, their accuracy averaged 95.6%, compared with LoRA-large's 94.7%. The coverage-performance tradeoff across all evaluated confidence thresholds is summarized in Table V.

This indicates that confidence could identify predictions with a higher likelihood of error on a limited statistical level, as a relatively higher proportion of incorrect predictions occurred at lower confidence levels. However, some incorrect predictions also occurred at higher confidence levels. Probability calibration

and confidence-based selective prediction can provide insight into which predictions may be worth further investigation but cannot eliminate uncertainty. When paired with the LIME and lexical analyses above, these methods provide further insight into when incorrect predictions may occur, while the LIME and lexical analyses indicate which words or features may contribute to them.

VI. Discussion

Four main conclusions were reached in this study. Firstly, both text representation and task adaptation demonstrated substantial impact on performance in stress classification. The best-performing Word2Vec model attained an average F1-score of 0.731 ± 0.018 and the best-performing fixed transformer representation attained an average F1-score of 0.787 ± 0.000. Task adaptation using transformer models yielded additional gains in performance; the strongest MPNet configurations achieved average F1-scores between approximately 0.830 and 0.839 across the three seeds. This suggests that pretrained contextual text representations contain information that proves more valuable in stress classification than locally trained static embeddings. Additionally, task adaptation yields further gains in performance.

Secondly, classifier-head-only training in the configurations evaluated here did not consistently yield strong results; this does not establish that frozen representations themselves were inadequate. Full fine-tuning resulted in gains in both ROC-AUC and F1-score on all three transformer models. The size of gain, however, appeared to be dependent upon the model and was not consistently reflected across all metrics. MiniLM classifier-head-only training remained particularly weak (ROC-AUC 0.671), although the same frozen MiniLM with mean-pooled embeddings and XGBoost attained ROC-AUC 0.822. The feature-based pipeline used mean pooling, whereas the sequence-classification head read the first-token pooler; pooling and classifier-training choices therefore confound a representation-only explanation. The gains from full fine-tuning were smaller for MPNet and DistilBERT. It appears that performance should be measured on multiple metrics rather than solely by accuracy.

Thirdly, LoRA appears to provide a good trade-off between performance and number of parameters adapted. Across all models and configurations tested, LoRA adapted only between 0.17% and 1.31% of the model parameters. For MPNet, performance was comparable to that of full fine-tuning, while DistilBERT LoRA configurations achieved F1-scores very similar to full fine-tuning. In the matched ten-seed experiment using MPNet, LoRA-large attained an average F1-score of 0.847 versus 0.822 for full fine-tuning, and a ROC-AUC of 0.913 versus 0.894. Both differences were statistically significant after Holm correction. This yields more support for the claim that LoRA-large can outperform full fine-tuning under these training settings while updating only a small fraction of the model's parameters. For MiniLM, LoRA did not yield a higher average F1-score than full fine-tuning. Also, increasing the rank of the LoRA adaptation did not consistently yield superior performance across backbones. For these reasons, the capacity of LoRA should be determined based on performance on a validation set rather than by assuming larger ranks of adaptation will always yield better results.

Finally, aggregate performance metrics were insufficient for characterizing model reliability. The original selected fully fine-tuned MPNet model yielded roughly 2.20 times as many false positives as false negatives. Both LIME and lexical analysis appear to show that affective and self-referential language may lead to misclassification between text that directly expresses stress and text that merely mentions stress-related topics. Additionally, false predictions appeared to occur with lower confidence compared to true predictions. However, some false predictions did also occur with relatively high confidence. Temperature scaling appears to improve probability calibration while selective prediction appears to yield improved performance on predicted instances while decreasing the number of predictions made. Through error analysis, explainability, calibration, and selective prediction, the analysis provided insights into model behavior that would not have been possible with aggregate performance metrics alone.

VII. Conclusions, Limitations, and Future Work

A. Conclusions

This paper investigates several approaches to automated stress identification from Reddit text including feature models, pretrained transformer models, end-to-end transformer models, parameter-efficient fine-tuning, uncertainty calibration, and explainability. The results show an improvement in performance from the original Word2Vec models to fixed pretrained transformer models and then to transformer models fine-tuned for the task of stress identification. The best performance was achieved using models of the MPNet architecture fine-tuned for the stress identification task. MPNet LoRA-large (about 1.07% of parameters updated during training) showed an average ROC-AUC of 0.913 ± 0.003 and an average F1-score of 0.847 ± 0.005 across the ten seeds in a matched experiment. Full fine-tuning (which updates all parameters) showed an average ROC-AUC of 0.894 ± 0.016 and an average F1-score of 0.822 ± 0.011.

The LoRA results indicate that comparable performance can be achieved while updating a relatively small fraction of the model parameters. The matched ten-seed MPNet experiment further showed statistically significant improvements in F1-score and ROC-AUC for LoRA-large over full fine-tuning. However, high aggregate performance does not necessarily indicate the absence of systematic errors, overconfident wrong predictions, or difficulty distinguishing direct expressions of stress from text discussing stress-related topics. The results suggest that pretrained transformer models have promise for stress identification but also demonstrate the value of parameter efficiency, error analysis, explainability, and confidence analysis in addition to performance metrics.

B. Limitations

This study used one publicly available Reddit dataset and one training-validation split, so the results do not represent

performance across time, communities, populations, or datasets. The initial experiments used three random seeds; the follow-up comparison between full fine-tuning and LoRA-large used ten paired seeds. MPNet full fine-tuning selected the upper boundary of its learning-rate grid, while several other configurations selected lower boundaries; the comparisons do not establish superiority over all possible hyperparameter settings. Dreaddit labels indicate stress or stress-related content, not clinical determinations about the writer. This study involved no clinical providers or patients, and the models made no diagnoses or interventions.

Only three transformer backbones and one parameter-efficient fine-tuning method were explored. Trainable parameter counts and percentages measured parameter efficiency. Wall-clock training and evaluation time and peak GPU memory were measured for the empirical-reporting reruns, but these measurements are hardware-specific; energy usage and broader computational-cost analyses were not measured. LIME provided local explanations on a reproducible sample but does not necessarily provide a full picture of model predictions. Uncertainty was explored only through prediction confidence on held-out, in-distribution Dreaddit samples. Uncertainty due to distribution shift, label ambiguity, and model parameters was not assessed.

C. Future Work

Additional experiments on other online texts and stress datasets, using more model settings and random seeds, could better assess error patterns and performance stability. Using clinician-verified data could also enable further analysis of the link between annotated stress in text and clinically meaningful measures if such analysis is ethical and feasible. External and out-of-domain testing could assess generalization across language use, online communities, and populations.

Future work could compare LoRA with other parameter-efficient methods and investigate energy usage, storage, and computational efficiency across hardware and deployment settings. Further explanation studies could distinguish text indicating stress from text discussing stress. Improved uncertainty estimation and human-in-the-loop selective prediction could flag potentially incorrect or low-confidence predictions for verification. These directions could further investigate trustworthiness, efficiency, and generalization while distinguishing stress classification from clinical evaluation.

VIII. Ethics

This work explores Reddit text that may contain sensitive content related to stress or mental health. Experiments on the public Dreaddit dataset examined aggregate language patterns and model performance, not specific individuals. No attempt was made to locate or contact individuals or assess their mental health.

The task identifies text containing stress or stress-related content; it does not diagnose mental health. Dreaddit annotations describe text content, not the author's mental health state, an important distinction when considering errors in other implementations. False positives classify non-stress content as stress-related; false negatives miss stress-related content. For these reasons, such a model should not be utilized alone in mental health diagnosis, treatment, or any other high-stakes scenario without further human review and validation.

No new Reddit content was scraped; all experiments used the public Dreaddit dataset, without attempts to recover usernames or identify or re-identify authors.

IX. Reproducibility

All experiments used the released Dreaddit train/test split. After repeated training texts and training examples overlapping with the official test set were removed, the resulting 2,817 examples were divided using a stratified 85/15 split into 2,394 training examples and 423 validation examples. The official 715-example test set was left unchanged. The original experiments used seeds 42, 123, and 2026. For the matched full-fine-tuning and LoRA-large experiments, the following ten seeds were used: 42, 123, 2026, 7, 13, 101, 202, 404, 777, and 1729. Both methods used the same seeds for paired comparisons and retained the same data splits. Model configurations and checkpoints were selected based on validation performance, and the official test set was not used for model training, hyperparameter selection, checkpoint selection, or temperature fitting.

The model checkpoints, hyperparameter search spaces, LoRA configurations, training settings, and model-selection procedures are reported in Section IV. Transformer experiments were run in Google Colab on an NVIDIA A100-SXM4-40GB GPU using PyTorch 2.11.0 with CUDA 13.0 (cu130), Transformers 5.16.1, Sentence-Transformers 6.0.1, Datasets 4.8.4, PEFT 0.20.0, Accelerate 1.14.0, and PyArrow 23.0.1. An anonymized reproducibility artifact is available at https://github.com/nlp-artifact-review/stress-nlp-artifact-review. The repository includes the full experimental notebook, analysis pipeline, pinned Python environment, and dataset-access instructions. For verification, the repository includes a CPU-only script recalculating ten-seed statistics from archived predictions and a GPU script rerunning matched full fine-tuning and LoRA-large configurations. The original three-seed MPNet comparison can be rerun separately; validation sweeps, per-seed predictions, and statistical summaries are also archived.

X. AI Use

Used OpenAI ChatGPT (GPT-5.6 Sol) only for help with R code translating the Python analysis pipeline's output into ggplots, some coding and debugging support, and LaTeX formatting. All generated code was run, and the outputs were compared with the actual experimental results before being used. No AI was used in study design, choosing follow-up analyses, analyzing results, drawing conclusions, generating research questions, or writing the manuscript.

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